

Electronic Engineering

Comprehensive Guide

Professional Edition 2026

Harz Digital Services

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Chapter 1

Introduction to Electronic Engineering

Electronic engineering is a discipline within electrical engineering that focuses on the design and application of electronic circuits, components, and systems. While electrical engineering deals with power generation and transmission, electronic engineering works with signals, data, and information processing at the component level.

Electronics vs Electrical Engineering

Electrical

Core Areas

Analog E

Key Components

Resistors
antennas

The Electronics Revolution

From vac
billions o

Chapter 2

Semiconductor Physics and Devices

Semiconductor devices are the building blocks of all modern electronics. Understanding their physics is essential for electronic engineering.

Semiconductor Basics

Semiconductors are materials with electrical conductivity between conductors and insulators. They are the backbone of modern electronics.

Doping

N-type: Doped with donor atoms (e.g., Phosphorus in Silicon), which provide free electrons for conduction.

PN Junction

When P-type and N-type semiconductors are joined, a depletion region forms at the junction. This region acts as an insulator, preventing current flow in one direction.

Diodes

Rectifier: A diode that allows current to flow in one direction only, converting AC to DC.

Transistors

BJT (Bipolar Junction Transistor): A three-terminal device. It can amplify signals and switch electronic circuits.

A BJT has $\beta = 100$ and base current $I_B = 20\mu A$. Collector current $I_C = \beta \times I_B = 100 \times 20\mu A = 2mA$.

Semiconductor Manufacturing

Process: The steps involved in creating semiconductor devices, from wafer fabrication to packaging.

Moore's Law

Transistor count in integrated circuits doubles approximately every two years, driving the growth of modern computing.

Chapter 3

Analog Electronics

Analog electronics deals with continuous-time signals. Despite the digital revolution, analog circuits remain essential for interfacing with the physical world.

Operational Amplifiers (Op-Amps)

The op-a
output im

Inverting amplifier with $R_{in}=10k$, $R_f=100k$. Gain = $-100k/10k = -10$. If $V_{in}=0.5V$, $V_{out} = -5V$.

Filters

Low-pass

Oscillators

Oscillator
phase sh

Amplifiers

Class A:

Applications

Audio an
front-end

Chapter 4

Digital Electronics

Digital electronics uses discrete signal levels (typically 0 and 1) to process and store information. It is the foundation of all modern computing.

Number Systems

Binary (b

Logic Gates

AND: Ou

Boolean Algebra

De Morga

Combinational Logic

Multiplex

Sequential Logic

Flip-flops

A 4-bit counter counts 0000 to 1111 (0 to 15) then wraps. With a 1MHz clock, it completes one cycle in 16 microseconds.

Chapter 5

Microcontrollers and Embedded Systems

Embedded systems combine hardware and software for dedicated functions. They are in everything from microwave ovens to spacecraft.

Microcontroller Architecture

A microc
families:

Programming Microcontrollers

Languag

Real-Time Operating Systems (RTOS)

RTOS m

Interfacing

Sensors:

Power Management

Battery-p
duty cycl

An ESP32 draws 240mA active, 20uA in deep sleep. A sensor reading every 10 seconds with 100ms active time gives 99% sleep, average current ~2.6mA. A 2000mAh battery lasts ~770 hours (32 days).

Chapter 6

Communication Systems

Communication systems transmit information from source to destination. They are the backbone of modern connectivity.

Modulation Techniques

Amplitude

16-QAM transmits 4 bits per symbol. At 1 Msymbol/s, data rate = 4 Mbps.

Multiplexing

FDM (Fre

Wireless Standards

WiFi (IEE

Shannon-Hartley Theorem

Channel

A channel with 1 MHz bandwidth and $S/N = 255$ (24 dB). $C = 1,000,000 \times \log_2(256) = 8$ Mbps.

Antennas

Dipole: S

Chapter 7

VLSI and PCB Design

Very Large Scale Integration (VLSI) and PCB (Printed Circuit Board) design are the physical implementation of electronic systems.

VLSI Design Flow

1. Specific

HDLs (Hardware Description Languages)

Verilog: C

PCB Design

Schemat

Signal Integrity

High-spe
(electrom

EDA Tools

Open sou

Manufacturing

PCB fabr

A 4-layer PCB stackup: Signal-Ground-Power-Signal. Ground plane provides low impedance return path and shielding.

Chapter 8

IoT and Emerging Technologies

The Internet of Things (IoT) connects physical devices to the internet, enabling smart homes, cities, industries, and healthcare.

IoT Architecture

Perception

IoT Protocols

MQTT: L

A smart farm uses LoRa sensors (soil moisture, temperature) sending data every 15 minutes via MQTT to a cloud dashboard. Alerts trigger when soil moisture drops below 30%.

Edge Computing

Process
operation

AI in Electronics

TinyML: L

Wearable Electronics

Smartwa
SpO2), B

Automotive Electronics

ECUs (E

Future Trends

Quantum

Chapter 9

Professional Practice and Standards

Electronic engineering professionals must understand standards, testing, and career paths to succeed in industry.

Key Standards

IEEE 802

Testing and Measurement

Oscillosc

ESD Protection

Electrosta
ESD-safe

Design for Manufacturing (DFM)

DFM gui
point acc

Career Paths

Hardware

Certifications

IPC certifi

Continuing Education

Electroni

